Extra_Assignment

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```
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, eval = TRUE, message = FALSE)
```

Here are the steps for us to use Bayesian approach

1.Define the question. Here, to estimate the potential revenue, we need to estimate θ because Revenue = 10000 + 5000* θ

2.Build the indicator. convert each day count into a binary outcome: exceed(=1) if count >1000,otherwise 0

3.Specify the likelihood(data model) Given p, the number of exceedance days X in n days follows a Binomial: X|θ ~ Binomial(n,p)

4.Choose a prior for p. Use Jeffreys prior for a binomial θ ~ Beta(0.5,0.5)

5.Update to the posterior conjugacy gives a Beta posterior θ| x~ Beta(0.5+x,0.5+n-x)

6.Summarize the posterior of θ

```
library(tidyverse)
library(kableExtra)


threshold  <- 1000
base_pay   <- 10000
bonus_pay  <- 5000
cred_level <- 0.95
alpha0     <- 0.5
beta0      <- 0.5


pedestrians <- read.csv("Data/pedestrians.csv", check.names = FALSE)


dbetabinom <- function(k, n, a, b) {
  choose(n, k) * base::beta(k + a, n - k + b) / base::beta(a, b)
}

posterior_params <- function(x, n, a0 = alpha0, b0 = beta0) {
  list(alpha = a0 + x, beta = b0 + (n - x))
}

posterior_summary_p <- function(x, n, a0 = alpha0, b0 = beta0, level = cred_level) {
  post <- posterior_params(x, n, a0, b0)
```

```
  post <- posterior_params(x, n, a0, b0)
  mean_p <- post$alpha / (post$alpha + post$beta)
  ci <- qbeta(c((1 - level) / 2, 1 - (1 - level) / 2), post$alpha, post$beta)
  list(mean = mean_p, ci = ci, alpha = post$alpha, beta = post$beta)
}

predictive_summary <- function(m, post_alpha, post_beta, level = cred_level) {
  mean_x <- m * post_alpha / (post_alpha + post_beta)
  ks <- 0:m
  pmf <- dbetabinom(ks, m, post_alpha, post_beta)
  cdf <- cumsum(pmf)
  low  <- ks[which(cdf >= (1 - level) / 2)][1]
  high <- ks[which(cdf >= 1 - (1 - level) / 2)][1]
  list(mean = mean_x, ci = c(low, high))
}

nms <- tolower(names(pedestrians))
is_long <- ("place" %in% nms | "location" %in% nms) &&
           ("number" %in% nms | "count" %in% nms | "pedestrians" %in% nms)

if (is_long) {

  place_col  <- if ("place" %in% nms) "place" else "location"
  number_col <- if ("number" %in% nms) "number" else if ("count" %in% nms) "count" els

  dat <- pedestrians %>%
    transmute(
      place  = .data[[place_col]],
      number = suppressWarnings(as.numeric(.data[[number_col]]))
    ) %>%
    filter(!is.na(place), !is.na(number)) %>%
    mutate(success = as.integer(number > threshold))

  summary_df <- dat %>%
    group_by(place) %>%
    summarise(n = n(),
              x = sum(success),
              .groups = "drop")

} else {

  numeric_cols <- pedestrians %>% select(where(is.numeric))


  long <- numeric_cols %>%
    mutate(row_id = row_number()) %>%
    pivot_longer(-row_id, names_to = "place", values_to = "number") %>%
    filter(!is.na(number))

  dat <- long %>%
```

```
    mutate(success = as.integer(number > threshold))

  summary_df <- dat %>%
    group_by(place) %>%
    summarise(n = n(),
              x = sum(success),
              .groups = "drop")
}


results <- summary_df %>%
  rowwise() %>%
  mutate(

    post_alpha = alpha0 + x,
    post_beta  = beta0  + (n - x),
    post_mean_p = post_alpha / (post_alpha + post_beta),
    p_ci_lo = qbeta((1 - cred_level) / 2, post_alpha, post_beta),
    p_ci_hi = qbeta(1 - (1 - cred_level) / 2, post_alpha, post_beta),

    expected_revenue = base_pay + bonus_pay * post_mean_p,
    revenue_ci_lo    = base_pay + bonus_pay * p_ci_lo,
    revenue_ci_hi    = base_pay + bonus_pay * p_ci_hi
  ) %>%
  ungroup() %>%
  arrange(desc(expected_revenue))


results %>%
  mutate(
    `p_mean` = sprintf("%.3f", post_mean_p),
    `p_CI`   = sprintf("[%.3f, %.3f]", p_ci_lo, p_ci_hi),
    `Revenue_mean` = sprintf("$%0.2f", expected_revenue),
    `Revenue_CI`   = sprintf("[$%0.2f, $%0.2f]", revenue_ci_lo, revenue_ci_hi)
  ) %>%
  select(place, n, x, p_mean = `p_mean`, p_CI = `p_CI`,
         Revenue_mean, Revenue_CI) %>%
  kbl(align = "r", caption = "Bayesian results by location (Jeffreys prior, threshold :
  kable_classic(full_width = FALSE)
```

Bayesian results by location (Jeffreys prior, threshold > 1000)

place	n	x	p_mean	p_CI	Revenue_mean	Revenue_CI
qv_melbourne	97	97	0.995	[0.975, 1.000]	$14974.49	[$14872.51, $14999.97]
southern_cross	97	31	0.321	[0.233, 0.417]	$11607.14	[$11165.67, $12083.14]
flinders_street	97	22	0.230	[0.152, 0.317]	$11147.96	[$10761.35, $11586.79]

From the chart above, the outcome is very clear. We use bayesian approach here, and qv_melbourne's credit interval is [14872.57,14999.97]. The southern cross's credit interval is [11165.67,12083.14]. The Flinder street's credit interval is [10761.35,11586.79].These credit interval means that "a 95% posterior probability that the parameter lies within the interval". However, in the group assignment, we use frequency approach, the confidence interval there means that "if we repeat the experiment countless times using the same sampling method, and each time calculate the confidence interval using the same method, then approximately 95% of the intervals will cover the fixed but unknown true parameter θ (e.g. the true proportion p).

In the group assignment, we use frequency approach. Here we use bayesian approach.There are difference between the two approaches in assumption.

- In frequency approach, the parameter is a solid number but unknown for us. Its randomness come from data sampling. However, in Bayes approach, these parameters are variables, they conform to a prior distribution before we sample, like the article above, we think P conform to a beta-binomial distribution

- In frequency approach, there is no prior test, all the outcomes come from data and sampling. But for bayesian approach, there is a prior test(here we use Jeffreys), and then we consider the data as a update to conclude the posterior.

- In frequency approach, we need to use point estimate, confidence interval, and test it. To make sure that the probability of the observations live in the CI is 95%. For bayes approach, it gives out the posterior distribution directly, A credible interval is "a 95% posterior probability that the parameter lies within the interval."

- Both of them require the data identically distributed and independent. Bayes requires more, the form of the prior distribution and its hyperparameters.

Relative strength(Bayesian)

- For Bayesian approach, It is better fit for a small sample, because of the existence of prior.

- Historical data can be used significantly if we choose bayesian approach. When there are multiple locations, the existing system Bayesian can be used to perform partial pooling to improve the stability of sorting

Relative weakness(Bayesian)

- It is not easy for us to get the priors. And a priori sensitivity is needed.

Relative strength(Frequentist)

- No need to prior test. When the sample is large enough, the outcome correspond to the population well.

Relative weakness(Frenquentist)

- When the sample is small, it is not stable.